7.1 二重积分的概念以及性质

2021年10月9日 8:31

一、单项选择题

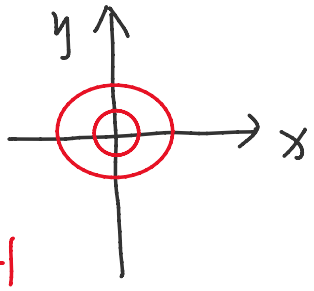
1. 设 $I=\iint\limits_D e^{x^2+y^2}\,d\sigma$，其中 D 为：$1\leqslant x^2+y^2\leqslant 4$，则下列结论正确的是 (C).

A. $3\pi e\leqslant I\leqslant 3\pi e^2$　　B. $3\pi e\leqslant I\leqslant 3\pi e^3$　　C. $3\pi e\leqslant I\leqslant 3\pi e^4$　　D. $3\pi e^2\leqslant I\leqslant 3\pi e^4$

m, M.

$S_D = 4\pi - 1\pi = 3\pi$

$e^{x^2+y^2}$



$r_1=1$

$r_2=2$

$1\le x^2+y^2\le 4$

$e^1\le e^{x^2+y^2}\le e^4$

$m=e$, $M=e^4$

$mS_D \le \iint\limits_D e^{x^2+y^2}dxdy \le MS_D$

$e(3\pi) \le \iint\limits_D \le e^4\cdot 3\pi$

$3\pi e\le I\le 3e^4\pi$

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2. 设 $I_1=\iint\limits_D (x+y)^3\,dxdy$，$I_2=\iint\limits_D \sin^3(x+y)\,dxdy$，其中 D 是由 $x=0$，$y=\frac{1}{4}$，$x=\frac{1}{2}$，$y=\frac{1}{2}$ 围成的封闭区域，则 I_1，I_2 之间的大小关系为 (B)

A. $I_1<I_2$　　B. $I_2<I_1$　　C. $I_1=I_2$　　D. I_1、I_2 大小无法确定

$f(x)\le g(x)$

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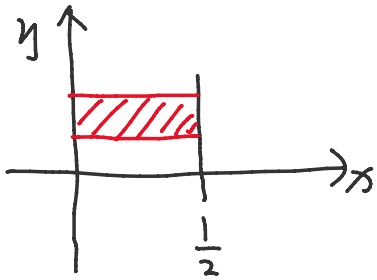


$[a,b]$　$f(x) \le g(x)$

$\int_a^b f(x)dx \le \int_a^b g(x)dx$

D.　$f(x,y) \le g(x,y)$

$\iint_D f(x,y)\,dxdy \le \iint_D g(x,y)\,dxdy$

$x>0$ 时，　$x > \sin x$

$\square > 0$ 时，　$\square > \sin\square$

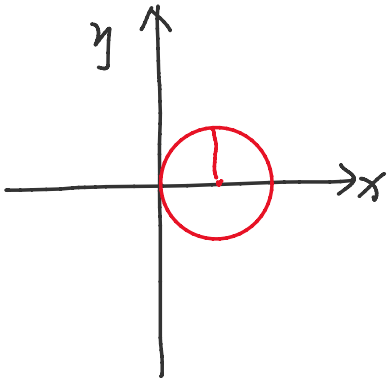
$x+y>0$，　$x+y > \sin(x+y)$

$(x+y)^3 > \sin^3(x+y)$

3. 设 $D = \{(x,y) \mid (x-1)^2 + y^2 \leqslant 1\}$，则 $\iint\limits_D dxdy =$　　(C)

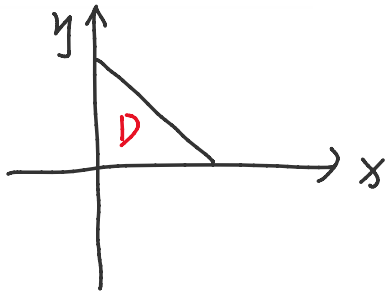
A. 3π　　B. 4π　　C. π　　D. π^2

圆心：$(1,0)$　半径 1.



$\iint\limits_D dxdy = S_D = r^2\pi = \pi$

2. 比较积分 $I_1=\iint\limits_D (x+y)^7 d\sigma$ 与 $I_2=\iint\limits_D (x+y)^8 d\sigma$ 的大小，其中 D 是由 Ox 轴、Oy 轴及直线 $x+y=1$ 围成的区域，则_____.



$0\le x+y\le 1$

$0<x<1$

$x^2>x^3>x^4$

$0\le (x+y)\le 1$

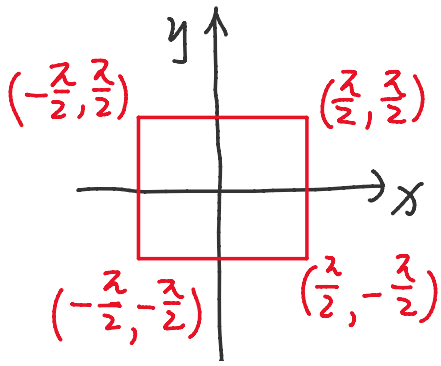
$(x+y)^7\ge (x+y)^8$

$\iint\limits_D (x+y)^7 dxdy \ge \iint\limits_D (x+y)^8 dxdy$

3. 积分 $I=\iint\limits_D |\cos^8 x\cos 9y|\, d\sigma$，其中 $D=\left\{(x,y)\,\middle|\,-\frac{\pi}{2}\leqslant x\leqslant\frac{\pi}{2},-\frac{\pi}{2}\leqslant y\leqslant\frac{\pi}{2}\right\}$，则 I 的取值范围是_____.

$|\cos\ \cos|$

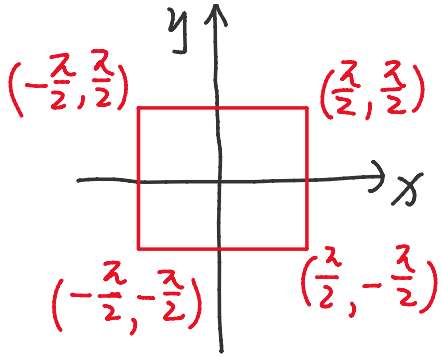
$mS_D\le \iint\limits_D f(x,y)dxdy\le MS_D$



$S_D: \pi^2$

$m: 0 \quad (\frac{\pi}{2},\frac{\pi}{2})$

$|\cos\ \cos|$ $\quad m\,\sigma_D = \iint_D |\cos x|\,dx\,dy = \cdots$



$S_D : \pi^2$

$m : 0 \quad (\frac{\pi}{2},\frac{\pi}{2})$

$M : 1 \quad (0,0)$

$$\boxed{0 \le I \le \pi^2}$$

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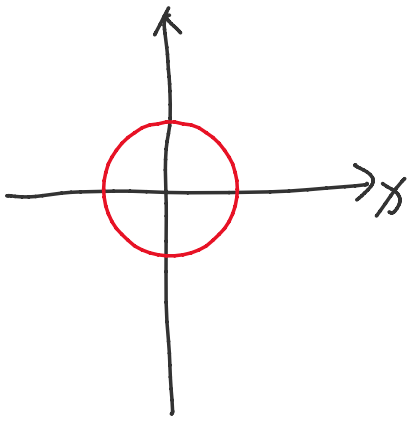
5. 设 $\iint\limits_{x^2+y^2\leqslant a} d\sigma = 16\pi$，则 $a =$ ___16___.

$$\iint\limits_{x^2+y^2\le a} dx\,dy = S_D = 16\pi$$



$x^2+y^2=r^2$

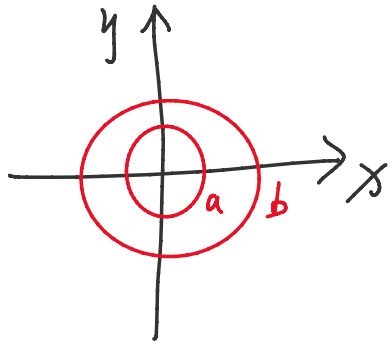
$r^2=a.$

$r=\sqrt{a}.$

$S_D = r^2\pi = a\pi = 16\pi \Rightarrow a=16$

7. 设$\iint\limits_D d\sigma = \pi$，其中 $D: a^2 \leqslant x^2 + y^2 \leqslant b^2$，这里 $a^2 + b^2 = 1$，则 $a =$ ___0___，$b =$ ___±1___.

$\iint\limits_D d\sigma = S_D = \pi.$



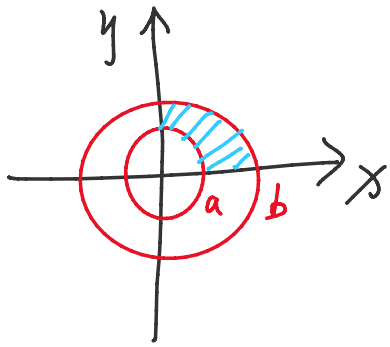
$S_D = b^2\pi - a^2\pi = (b^2 - a^2)\pi$

$= \pi$

$\Rightarrow \begin{cases} b^2 - a^2 = 1 \\ a^2 + b^2 = 1 \end{cases}$ $\quad 2b^2 = 2 \quad b^2 = 1 \quad a^2 = 0$

9. 设区域 D 为第一象限内的扇环域：$2 \leqslant x^2 + y^2 \leqslant 4$（含坐标轴），则二重积分 $I = \iint\limits_D \ln\sqrt{x^2 + y^2}\,dxdy$ 的值所在的范围为_____.

$mS_D \leq \iint\limits_D \leq MS_D$



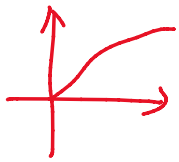
$S_D = \frac{1}{4}(4\pi - 2\pi) = \frac{\pi}{2}$

$2 \leq x^2 + y^2 \leq 4$

$\sqrt{2} \leq \sqrt{x^2+y^2} \leq \sqrt{4} = 2$

$a < b$

$e^a < e^b$



$\ln\sqrt{2} \leq \ln\sqrt{x^2+y^2} \leq \ln 2$

$\sqrt{a} < \sqrt{b}$

$\underbrace{\frac{1}{2}\ln 2}_{m} \leq \underbrace{\ln\sqrt{x^2+y^2}}_{} \leq \underbrace{\ln 2}_{M}$

$\frac{1}{2}\ln 2 \leq \iint \ln\sqrt{x^2+y^2}\,dxdy \leq \frac{1}{2}\ln 2$

$$\frac{1}{2}\ln 2\cdot\frac{1}{2}\pi \le \iint_D \ln\sqrt{x^2+y^2}\,dx\,dy \le \frac{1}{2}\pi\cdot\ln 2.$$

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